Customer Shopping Behavior — Overview

Analysis of 3,900 purchases to reveal spending, segments, product & subscription trends.



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Dataset Snapshot

Scale

Rows: 3,900 • Columns: 18

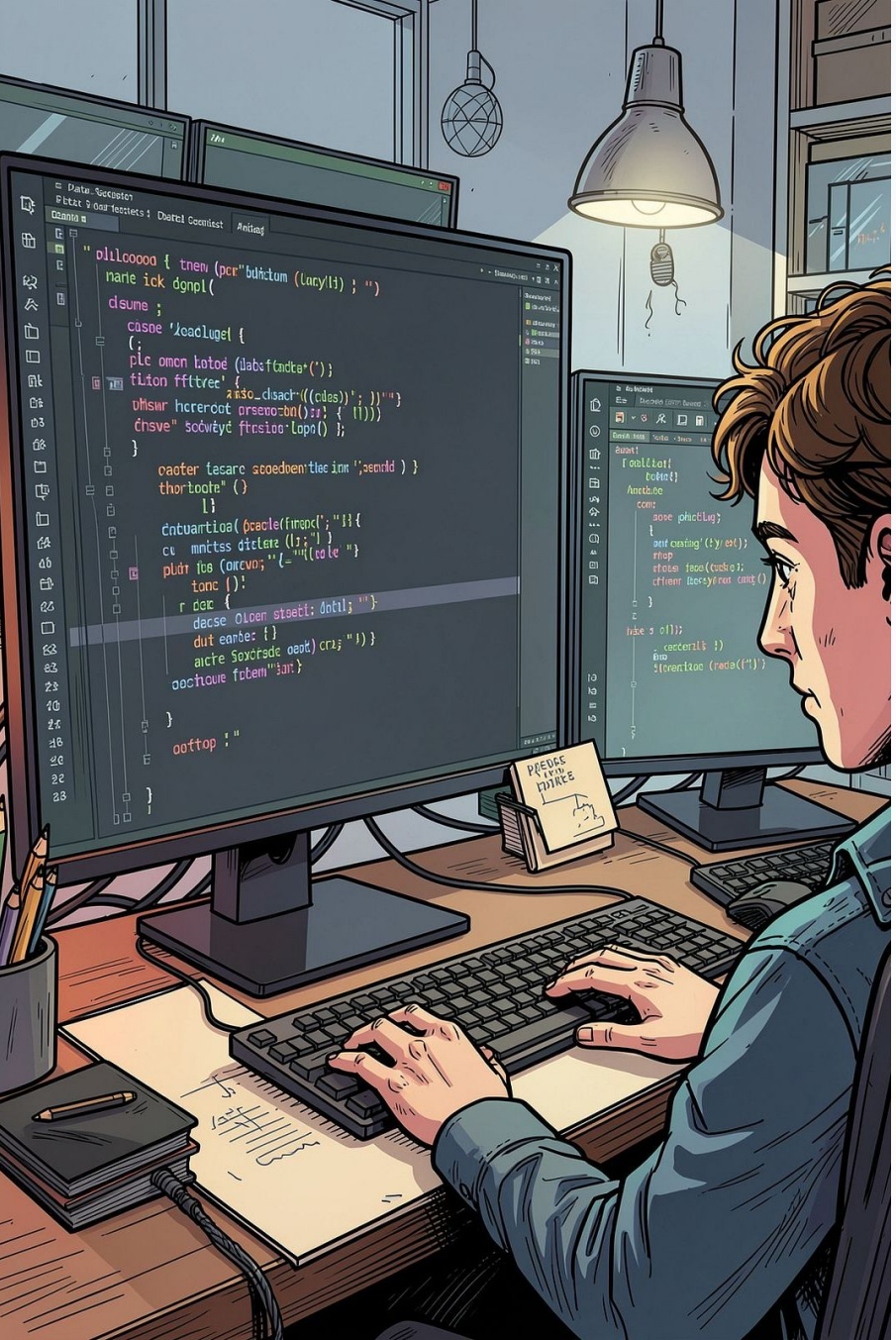
Key Features

Demographics, Purchase details, Behavior, Shipping

Missing Data

37 nulls in Review Rating



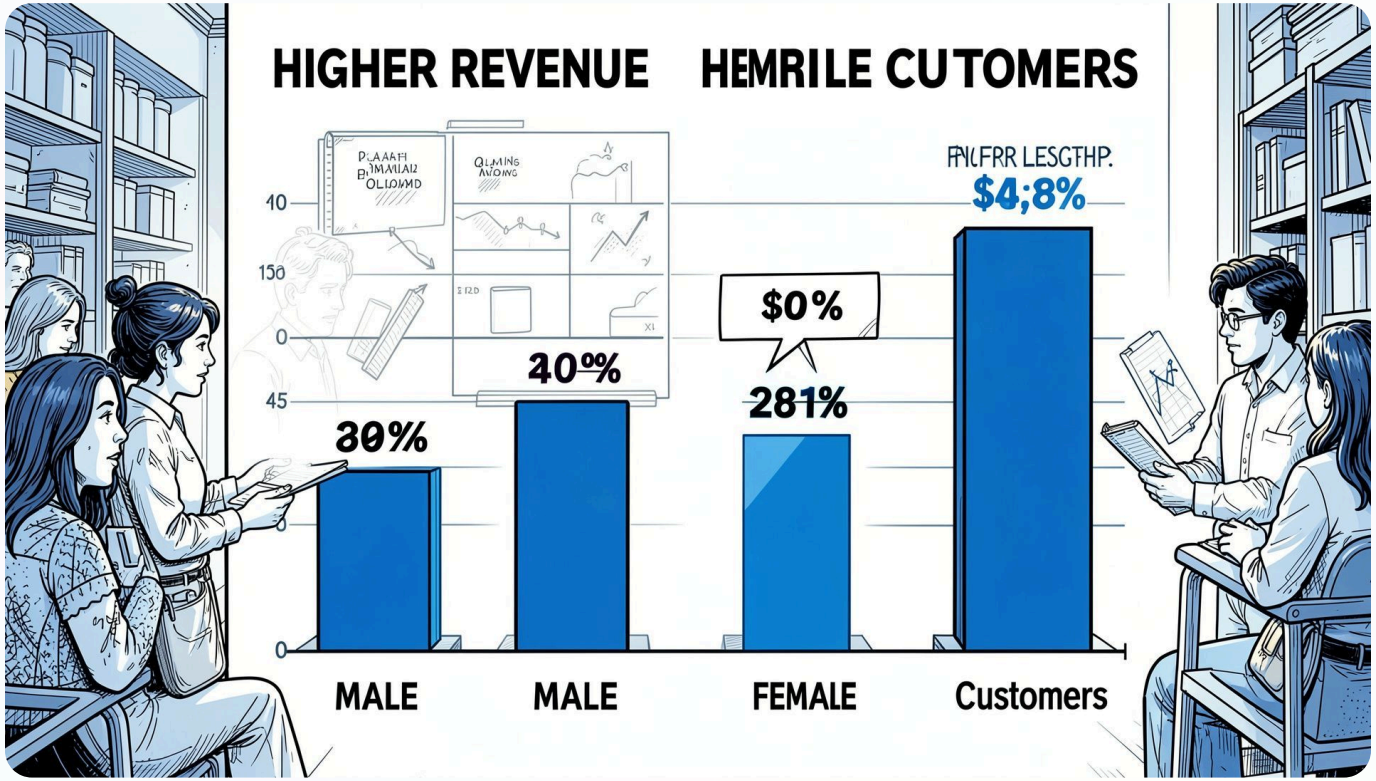


Python EDA & Preparation

- Loaded with pandas
- df.info() & .describe()
- Imputed Review Rating by category median
- Renamed to snake_case
- Engineered age_group & purchase_frequency_days
- Dropped redundant promo_code_used
- Exported cleaned DF to PostgreSQL

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Revenue by Customer Attributes



By Gender

Female: 75,191 • Male: 157,890

By Age Group

Young Adult: 62,143 • Middle-aged: 59,197 • Adult: 55,978 • Senior: 55,763

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Top Products & Ratings



Gloves

Avg Rating 3.86



Sandals

Avg Rating 3.84



Boots

Avg Rating 3.82

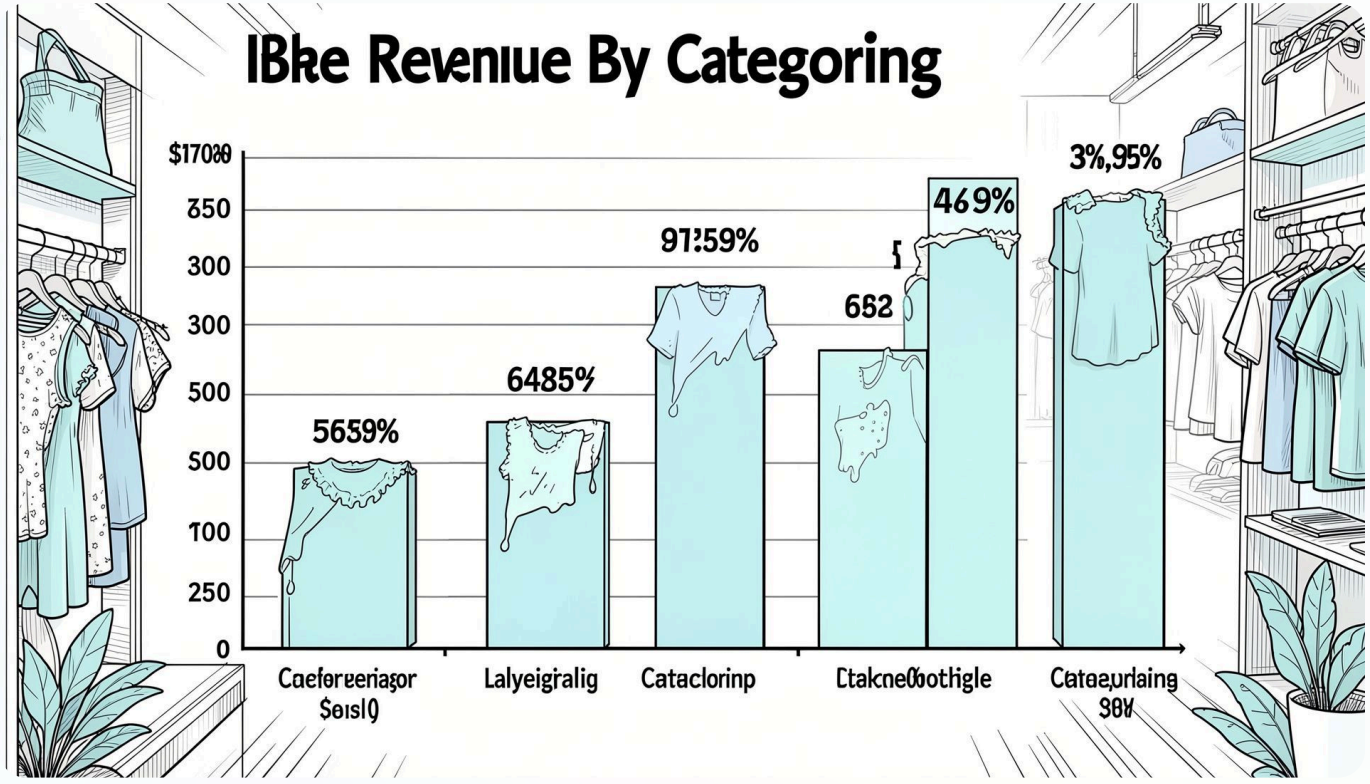
Top rated items guide product positioning and promotions.



Hat 3.80 • Skirt 3.78

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Category Performance



Revenue (approx)

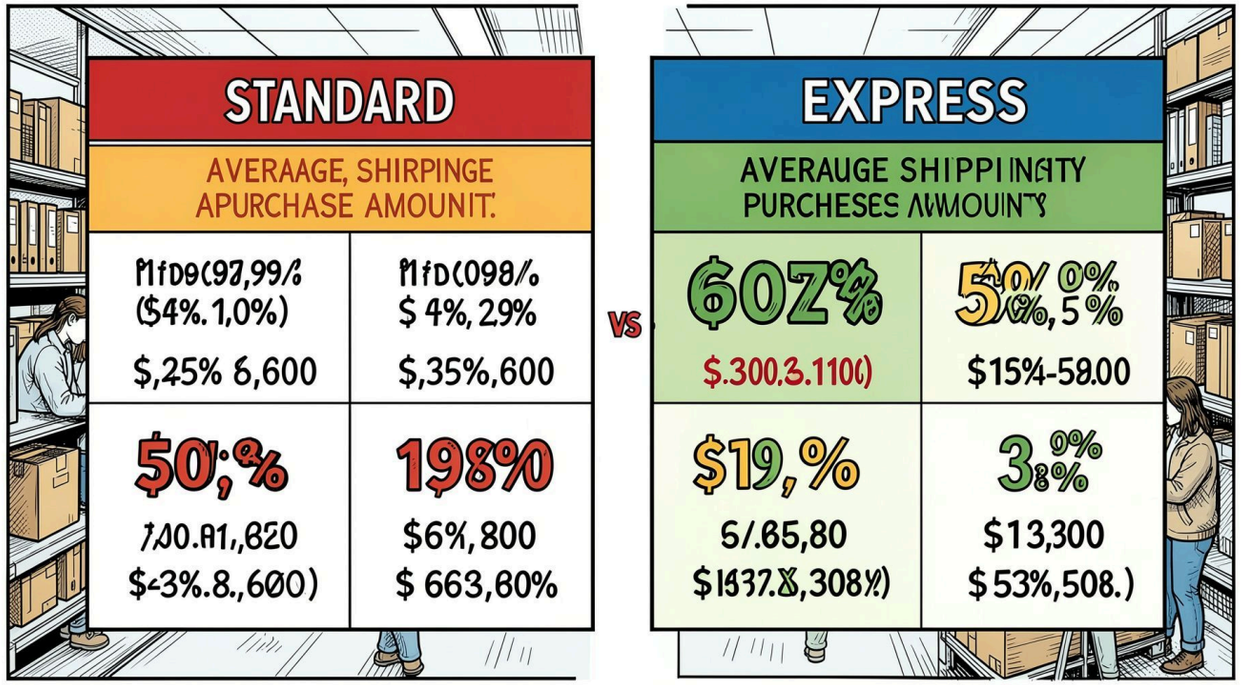
Clothing ~100K • Accessories ~75K • Footwear ~35K • Outerwear ~20K

Sales (units)

Clothing 1,700 • Accessories 1,250 • Footwear 600 • Outerwear 350

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Shipping & Discounts



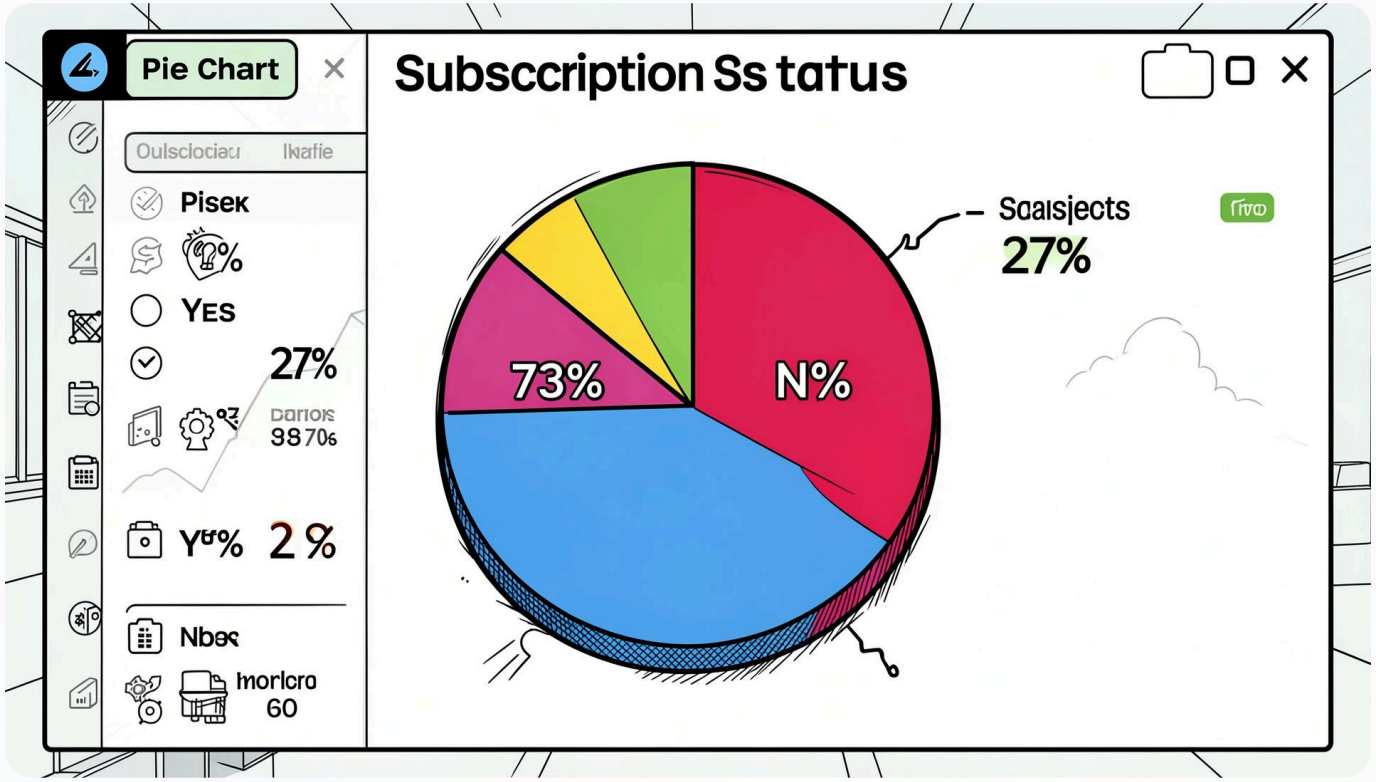
Standard avg: $58.46 • Express avg: $60.48

High-spending discount users identified (839 rows)



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Customer Segments & Subscriptions



Subscribers: 1,053 • Avg spend 59.49 • Total rev 62,645

Non-subscribers: 2,847 • Avg spend 59.87 • Total rev 170,436

Loyal

3,116 customers

Returning

701 customers

New

83 customers

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Discount-Dependent Products



High discount reliance suggests review of discount policy to protect margins.



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Recommendations & Dashboard



01

Boost Subscriptions

Promote exclusive subscriber benefits

02

Loyalty Programs

Reward repeat buyers to grow Loyal segment

03

Discount Policy

Balance promotions with margin control

04

Product Positioning

Highlight top-rated & best-selling items

05

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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